Before we begin...

Business Processes Modelling

An Introduction for Beginners

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Introduction to UML

What is UML?

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- English, Polish,... are natural languages
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- UML uses **graphical elements** to communicate
- You build **sentences** in UML using graphical elements

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- You build **sentences** in UML using graphical elements
- UML overcomes national language barriers
- UML saves hours spent in meetings

Grammar & Vocabulary

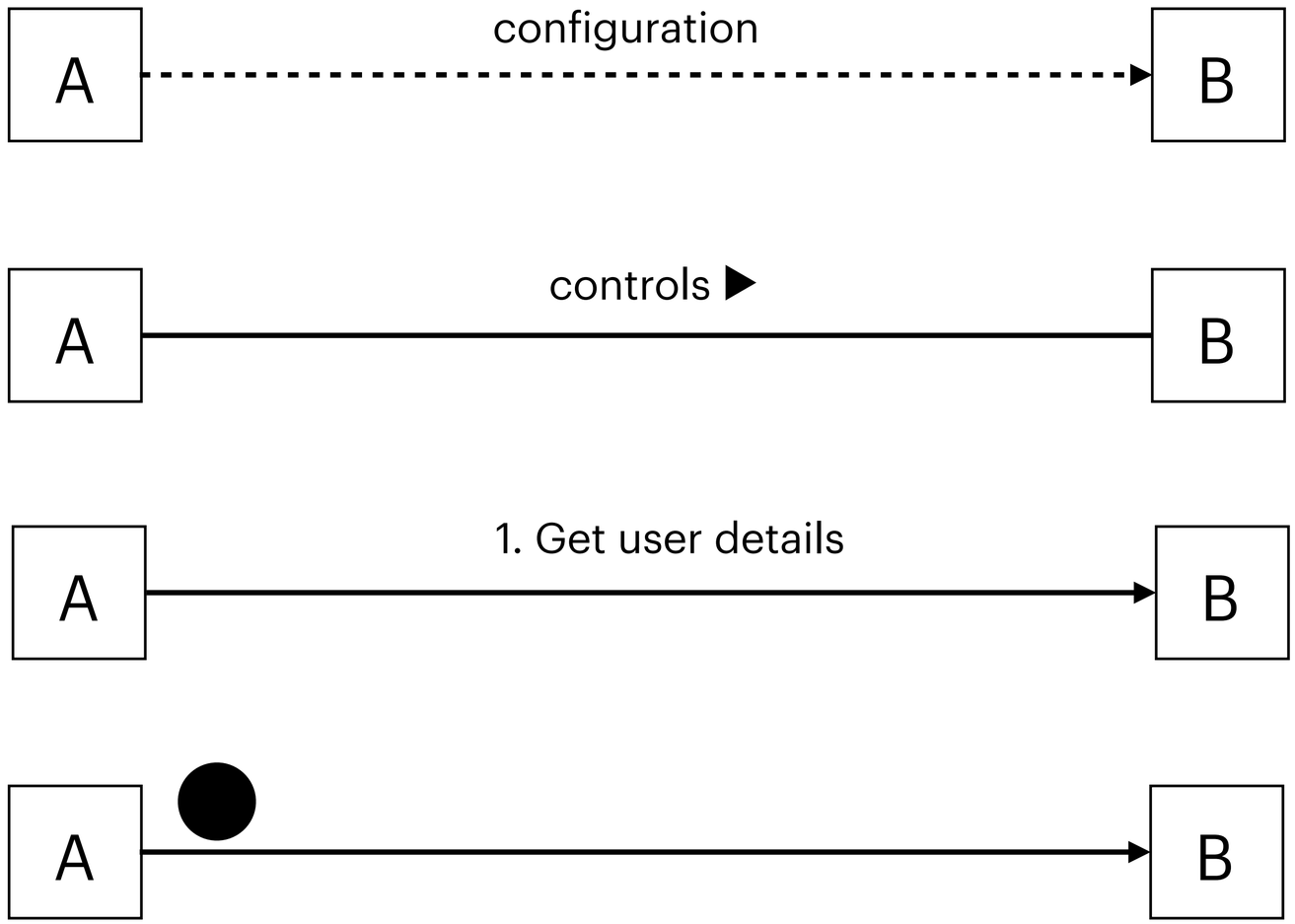
- UML is a **language** — has **grammar** & **vocabulary**
- UML is **graphical** language — vocabulary is **shapes** & **arcs**
- UML has **grammar** — answers how arcs connect shapes?

Grammar & Vocabulary

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- UML has **grammar** — answers how arcs connect shapes?
- UML has **semantics** — shapes & arcs have a **meaning (!!!)**
- UML conveys **information (!!!)** — using diagrams

Note *UML is a variant of Knowledge Graph*

Arcs (!!!)



Dependency

A depends on B, such that A obtains configuration from B.

Association

A is in relationship with B, such that A controls B.

Control Flow

A sends message to B, and B takes control, B continues, and then control returns to A.

Data Flow

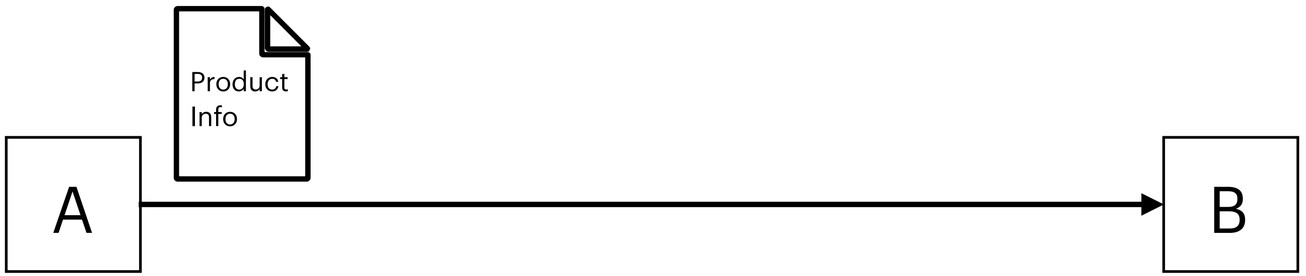
A sends data to B, and A stays in control, and B replies with status.

Arcs

Enhanced Data Arrows

Arcs

Enhanced Data Arrows



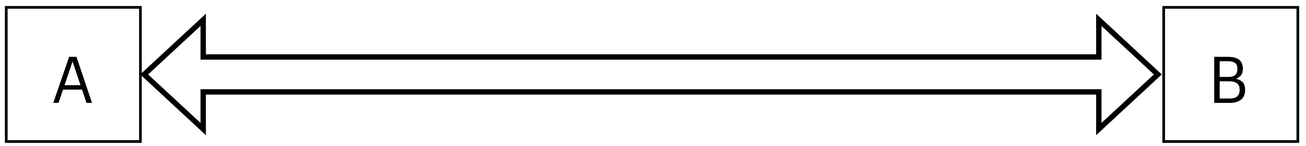
Data Flow

Data Flow arc is an arrow with an object



Thick Arrow

To emphasise data flow on some diagrams, thicker arrow can be often used



Fat Arrow

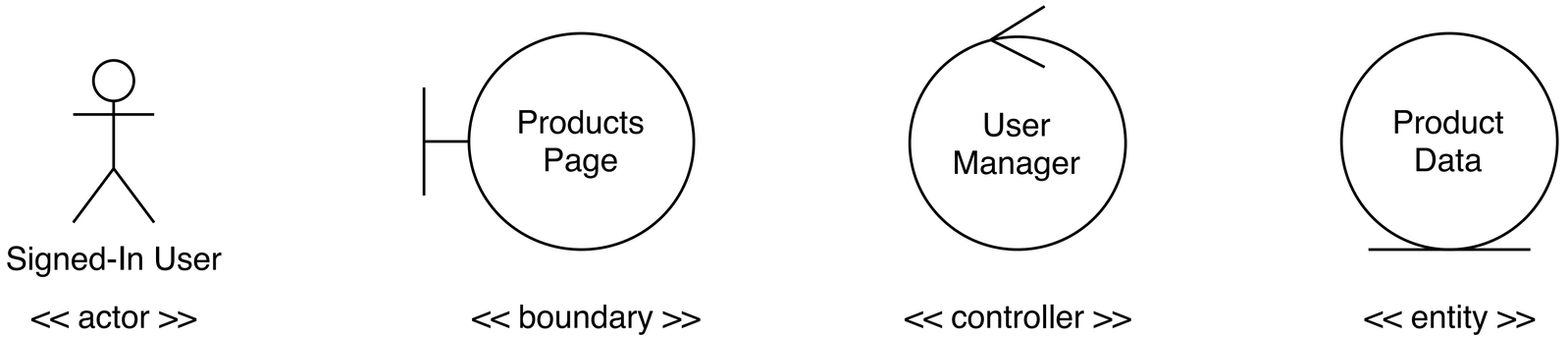
To emphasise data lanes on some diagrams, fat arrow can be used

Collaboration

Stereotypes

Collaboration

Stereotypes

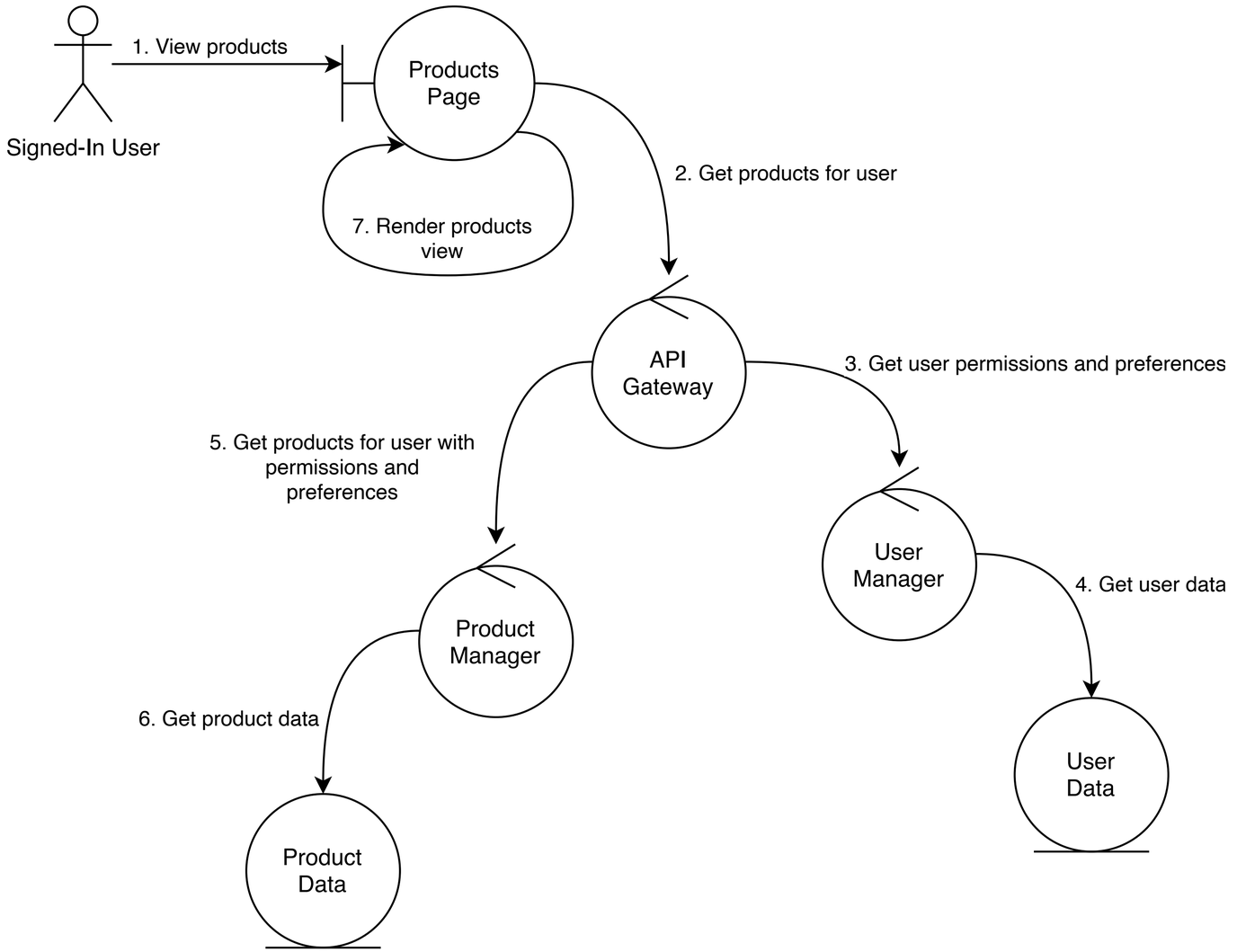


- << actor >> — activity initiator (human, service, ...)
- << boundary >> — communication interface (web page, web API, ...)
- << controller >> — active component (service, module, class instance...)
- << entity >> — static object (data record, resource, class instance, ...)

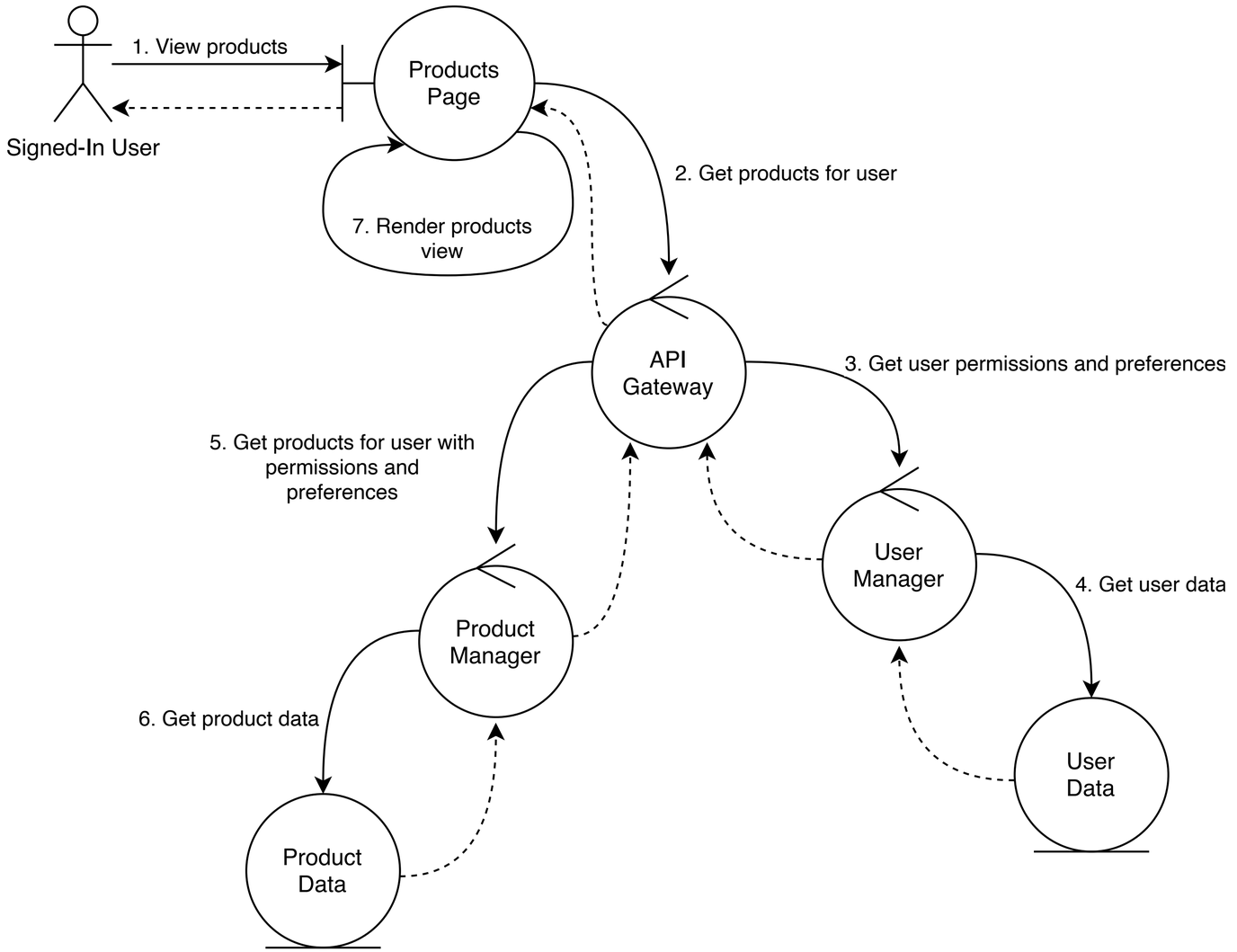
Collaboration

Diagram

Collaboration
Diagram



Collaboration
Diagram



Collaboration

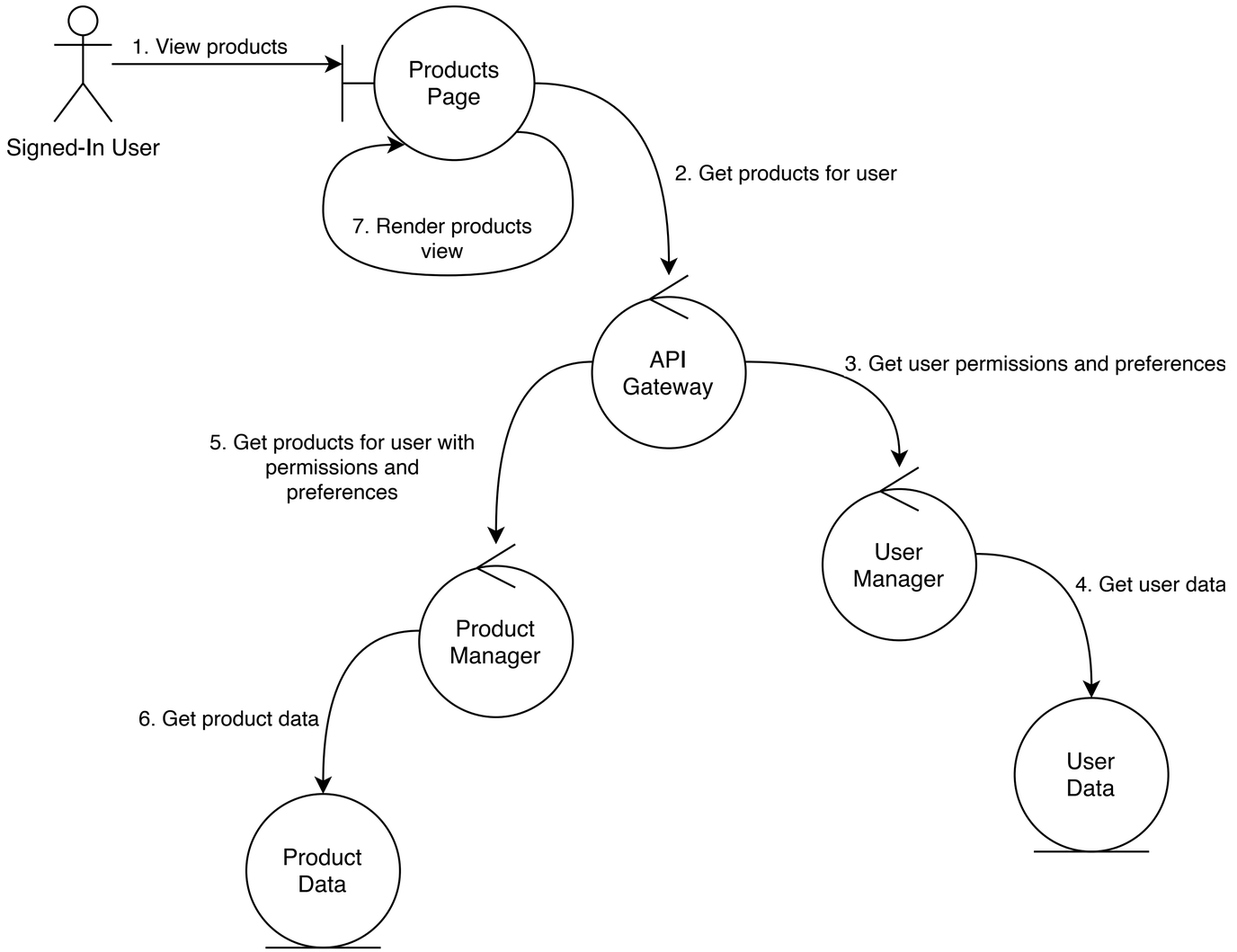
Diagram



WE DON'T NEED RETURN ARROWS!!!

(KISS & YAGNI)

Collaboration
Diagram

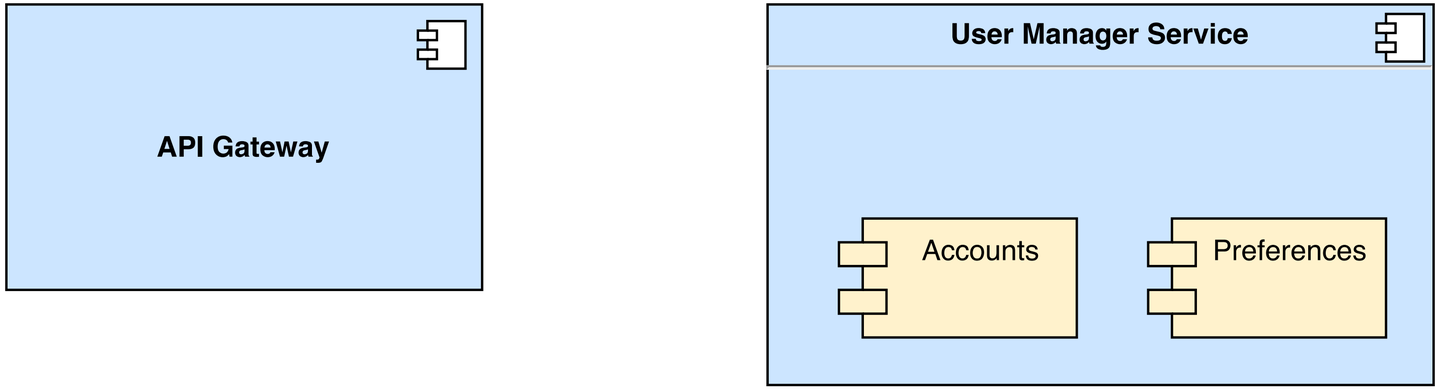


Component

Stereotypes

Component

Stereotypes

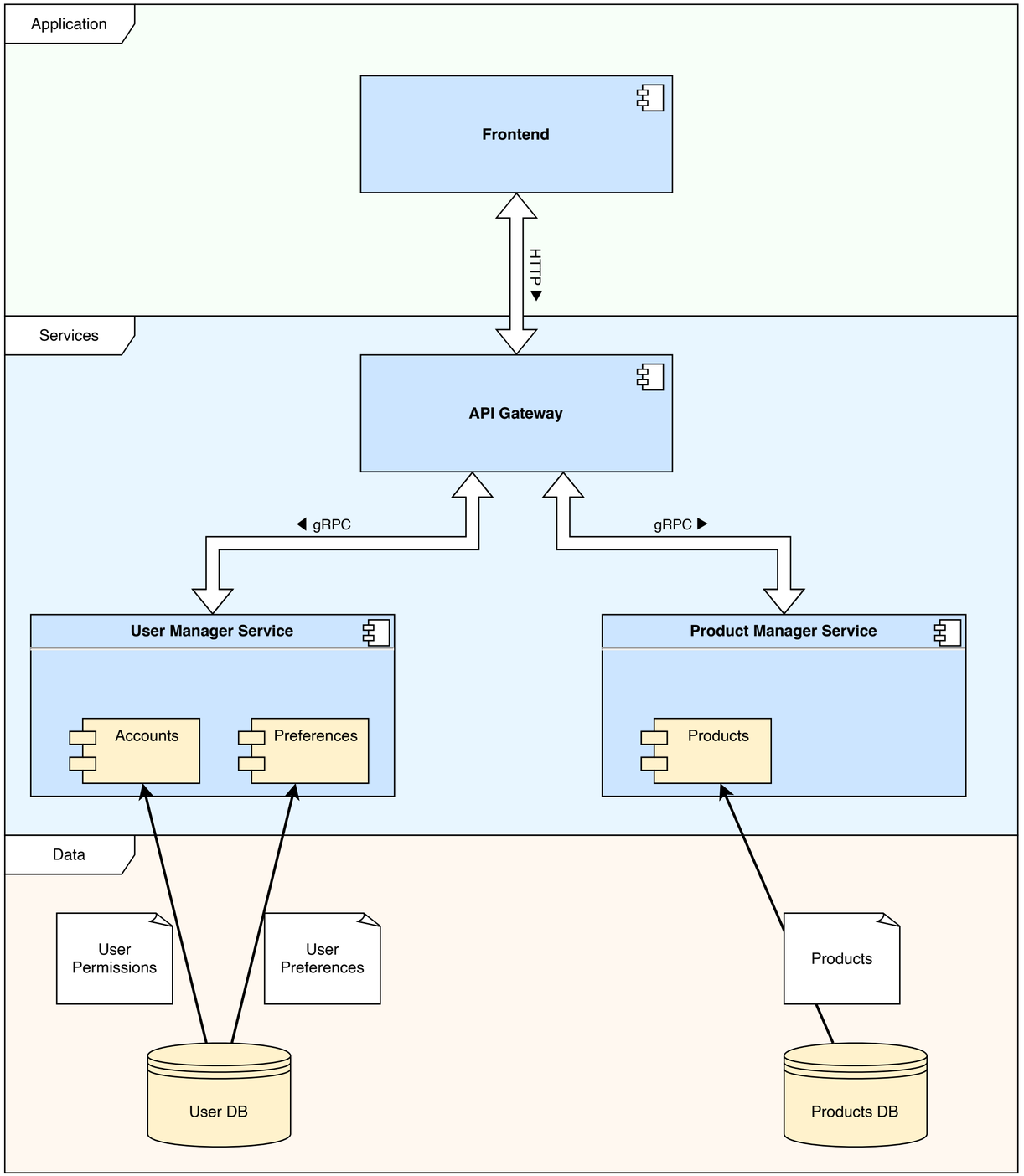


- << component >> — simple component, e.g. API Gateway
- << composite >> — composite component, e.g. User Manager Service
- << module >> — inner component, e.g. Accounts, Preferences

Component

Diagram

Component
Diagram



Activity

Stereotypes

Activity

Stereotypes

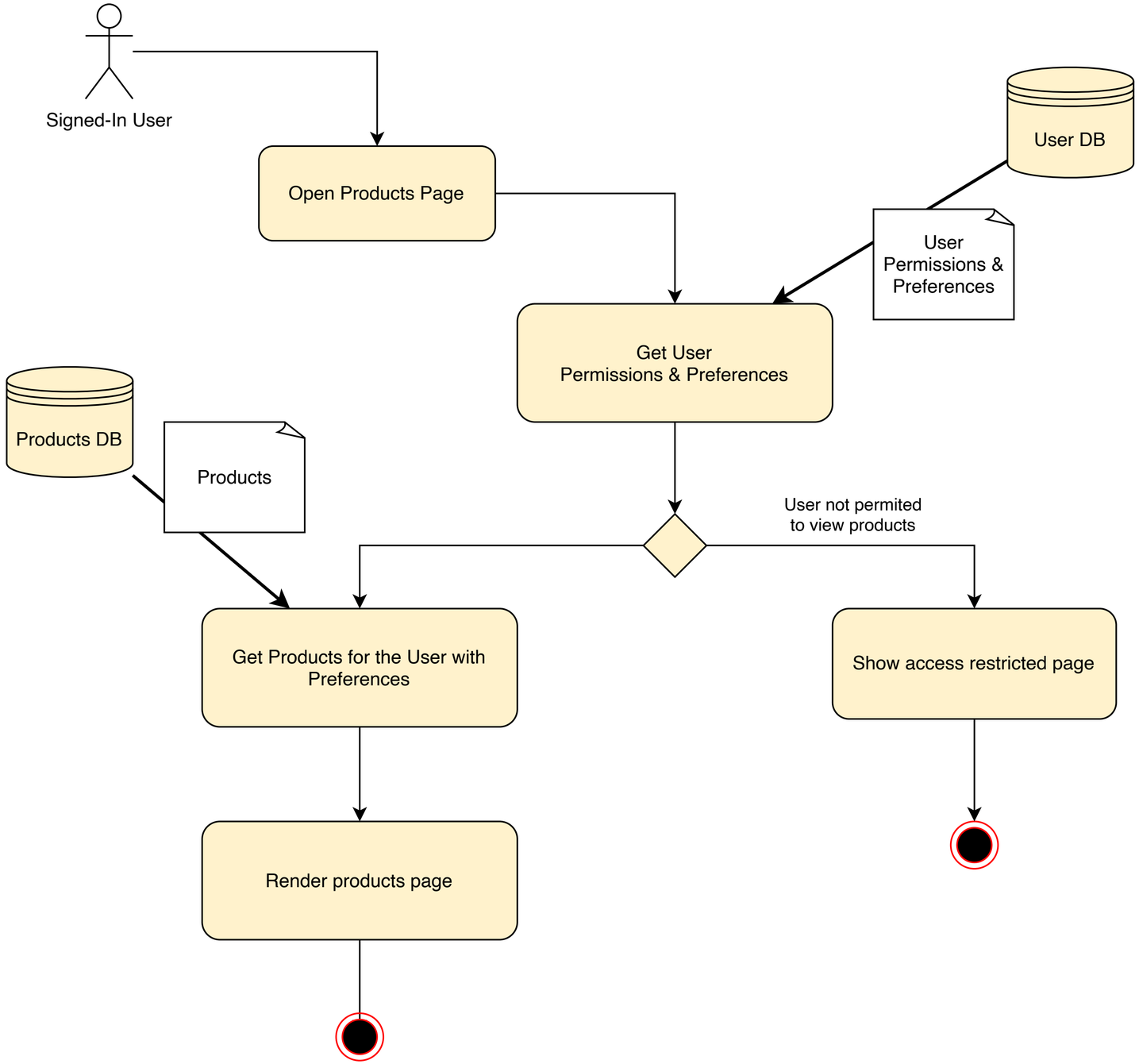


- << actor >> — initiator (human, service, ...)
- << activity >> — describes what happens
- << decision >> — a point where decision needs to be made
- << end >> — end of flow (can have multiple)

Activity

Diagram

Activity
Diagram

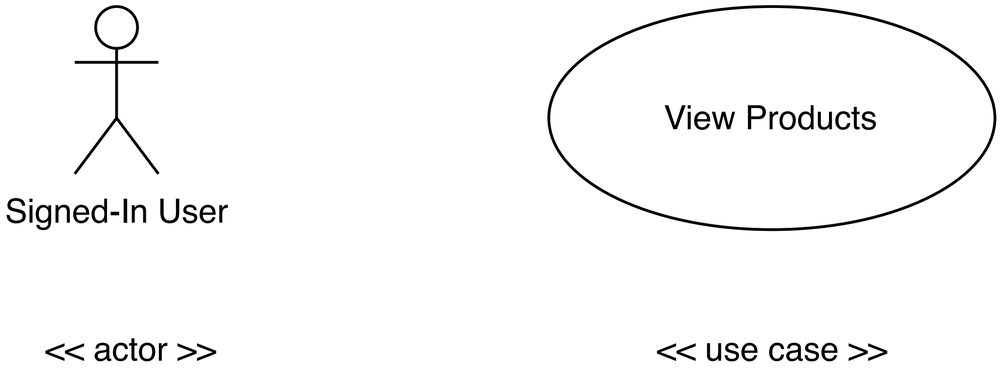


Use Case

Stereotypes

Use Case

Stereotypes

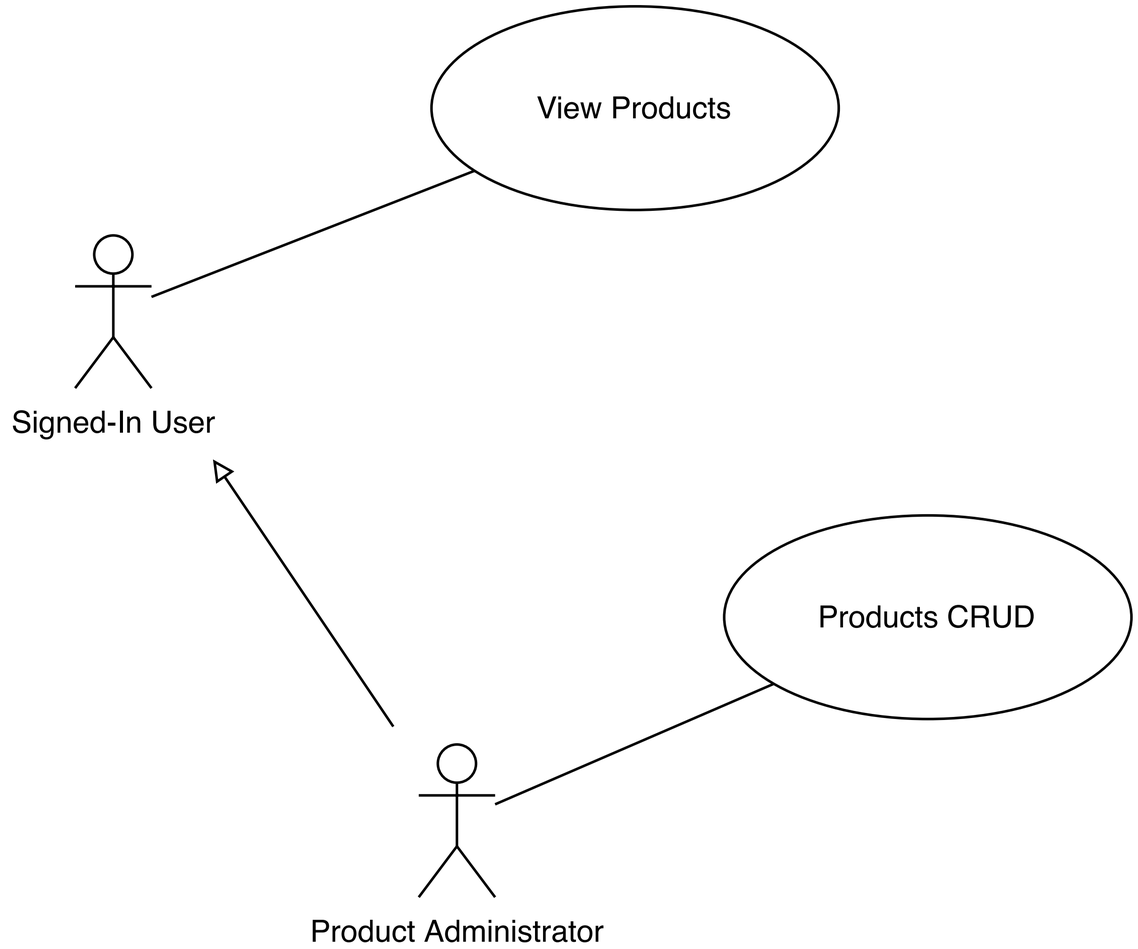


- << actor >> — receiver of the use case
- << use case >> — name of the use case

Use Case

Diagram

Use Case
Diagram



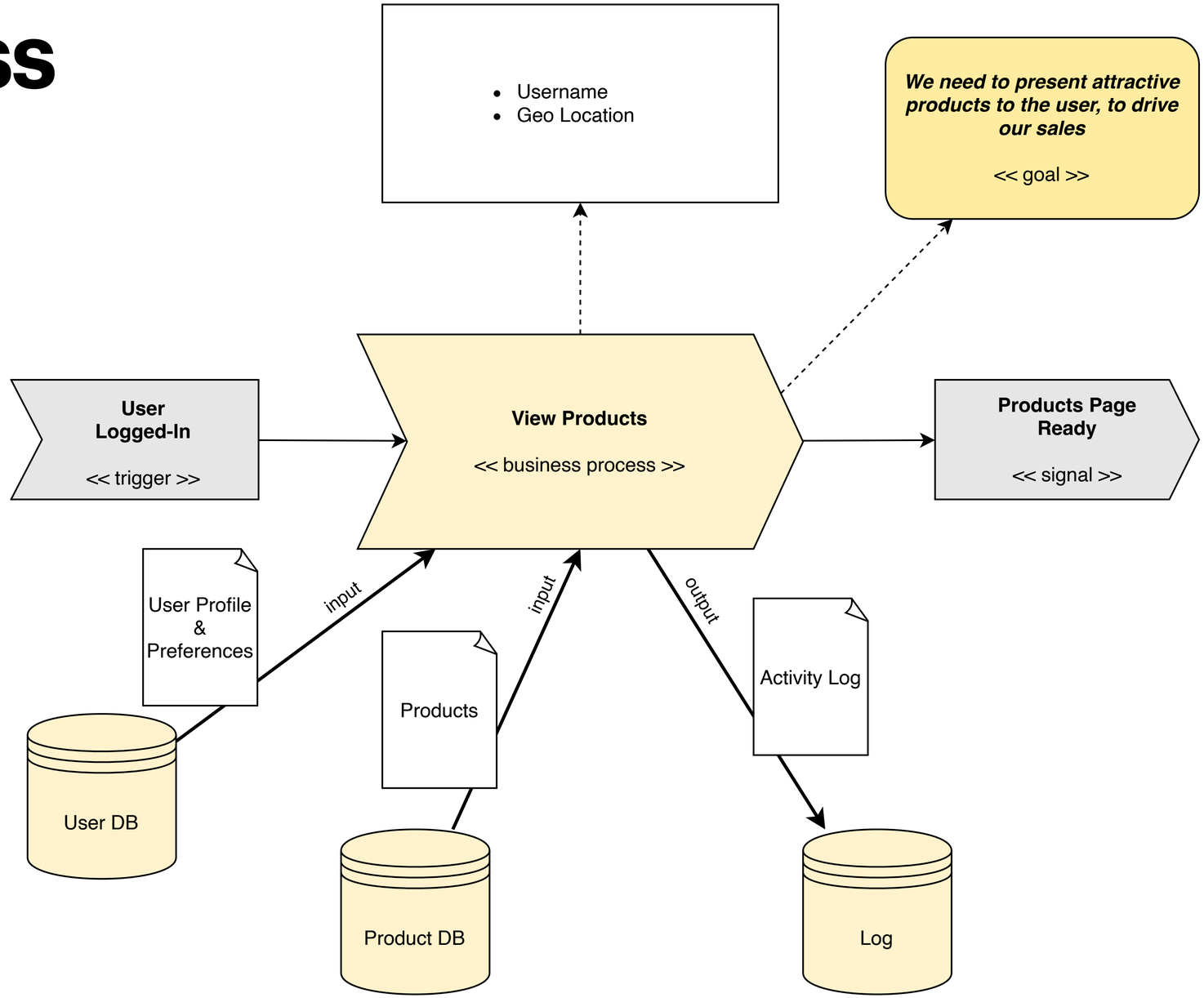
Business Process

Diagram

Business Process

Diagram

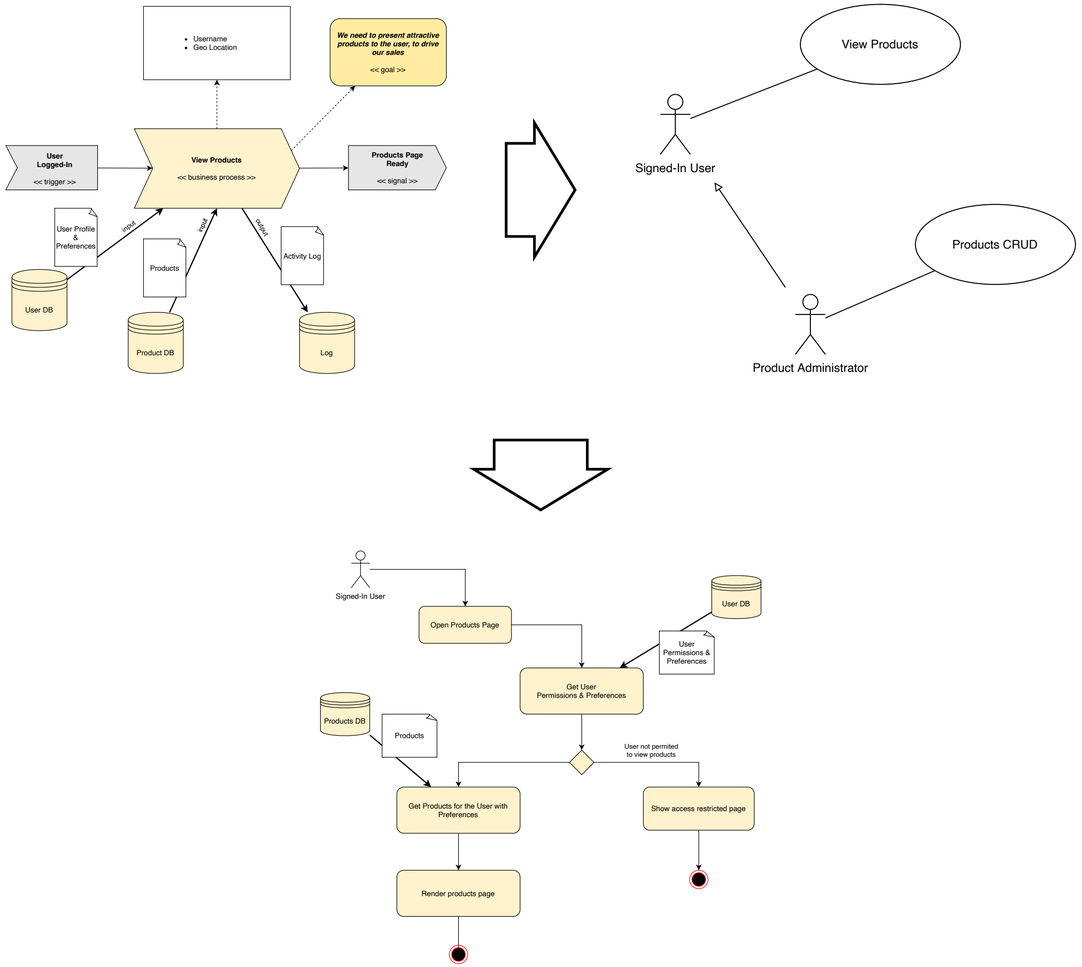
- **Name**
- **Goal**
- **Start Trigger**
- **End Signal**
- Input(s)
- Output(s)
- **Dependencies**



Business Processes Modelling

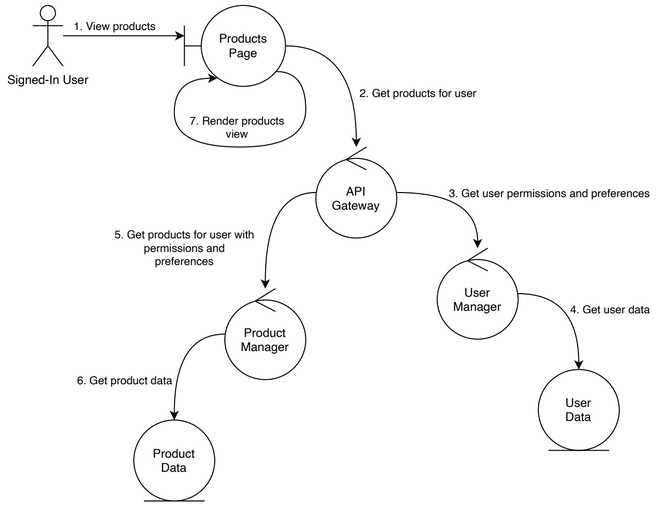
Analysis

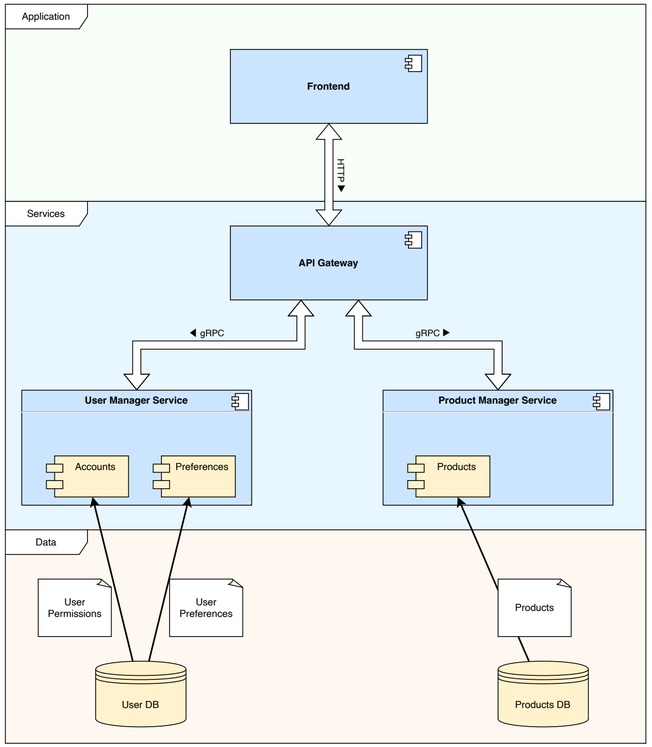
- Business Processes *(list)*
- **Business Process** *(Eriksson-Penker)*
- **Use Case**
- **Activity** *(BPMN)*
- Domain Model



Design

- Interaction
  - **Collaboration**
  - State
  - Sequence
- **Component**
- Deployment





Demo...

Q & A

Business Processes Modelling — An Introduction for Beginners

Project version: `0.2`

Project location: https://github.com/sadhbh-c0d3/cc-docs

Tool used for diagrams: https://draw.io

Thank you!

PUBLIC DOMAIN